

Vaibhav Yenare

St. Xavier's College (Empowered Autonomous), Fort, Mumbai, 400 001, India

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Education

St. Xavier's College (Empowered Autonomous), Mumbai

Master of Science in Physics (Specialisation in Astrophysics) | CGPA : 8.35/10

[Jul 2023 - Mar 2025]

St. Xavier's College (Empowered Autonomous), Mumbai

Bachelor of Science

Majors: Physics, Minors: Mathematics & Statistics | CGPA : 7.32/10

[Jun 2020 - April 2023]

Publications

Yenare, V., Chatterjee, S., Ganguly, A., Nayak, P. et al. (*in preparation*). "Tracing White Dwarf- Main Sequence Binaries using UV-Optical Data." To be submitted to *The Astrophysical Journal*.

Patel, C., Akbari, K., Nadar, R., **Yenare, V.**, Choudhury, M. et al. (*in preparation*). "Temporal and Spectral Evolution of 4U 1630-47 During the RXTE Era." To be submitted to *Journal of High-Energy Astrophysics (JHAP)*

Research Experience

Tata Institute of Fundamental Research (TIFR), Mumbai

Research Project (MSc Thesis), Supervisor : **Prof. Sourav Chatterjee.**

[Nov 2024 - Present]

- Implemented a **two-step SED fitting methodology** using VOSA to independently model Main Sequence and White Dwarf components in binary systems.
- Designed and built Python-based data pipelines to perform catalog cross-matching (GALEX GR6+7, Gaia DR3), photometric filtering, and flux correction.
- Identified over **10,000 WD-MS binary candidates**, creating the largest sample to date in which various systems are dominated by the main-sequence star's contribution.
- Currently preparing a first-author manuscript detailing these findings for submission to *The Astrophysical Journal* (ApJ). ([Thesis Report](#))

St. Xavier's College (Empowered Autonomous), Mumbai

Research Project, Supervisor : **Prof. Manojendu Choudhury**

[Jan 2025 - Present]

- Analysed RXTE archival data of 4U 1630-47 to study the joint evolution of timing and spectral properties throughout all the outburst cycles.
- Investigated Spearman rank correlations between key spectral-timing parameters, identifying statistically significant relations associated with QPO occurrences.

Technical Abilities

Programming : Python, BASH, SQL

Softwares : VOSA, HEASOFT, CASA, CARTA, CIAO, MESA, SAO-DS9

Python Modules : Astropy, mwddust, bagpipes, bibly, GWpy, Einsteinpy, OpenCV, Galpy, sklearn

Markup : LaTeX

Relevant Short Projects

Spin Estimation of M33 X-8 Black Hole by Spectrum Analysis | *St. Xavier's College, Mumbai*

- Analysed NuSTAR data of the ULX M33 X-8 using XSPEC to estimate the black hole's spin (~ 0.85) through spectral fitting with diskbb, simpl, and kerrbb models. Achieved an excellent fit ($\chi^2 \sim 1.09$), offering insights into accretion processes in stellar-mass black holes. ([Project Report](#))

Molecular spectral line analysis of SiS for IRC+10216 | *St. Xavier's College, Mumbai*

- Used VLA radio observations to extract spectral lines of SiS molecules in the circumstellar envelope of IRC+10216. Carried out data reduction, flagging, and calibration using CASA. Focused on understanding and applying standard interferometric procedures to obtain accurate spectral information. ([Project Report](#))

Photometric data analysis for estimating galaxy parameters | *St. Xavier's College, Mumbai*

- Used multi-band photometric data to estimate galaxy properties such as stellar mass, age, star formation rate, metallicity, and dust content. Applied stellar population synthesis models and Bayesian analysis for parameter estimation, uncertainty quantification, and model comparison. ([Project Report](#))

Luminosity Distance Estimation of BH-BH Mergers | *St. Xavier's College, Mumbai*

- Used the Bilby library to perform Bayesian inference on gravitational wave data from BBH mergers. Estimated key parameters such as mass ratio, chirp mass, and luminosity distance. Calculated Bayes factors to assess signal vs. noise likelihood. Results improve distance measurements relevant to cosmology. ([Project Report](#))

Probing AGN Activity in NGC 660 | *St. Xavier's College, Mumbai*

- Studied the recent radio outburst in the polar-ring galaxy NGC 660 (~ 13 Mpc) using e-MERLIN and EVN radio interferometry along with X-ray spectroscopy. Analysed the radio spectral energy distribution and high-resolution imaging of jet-like features, providing evidence for a transition into a low-luminosity AGN phase. ([Project Report](#))

Workshops

Open Data Workshop 2024

School: GW Open Data Workshop

Issued: April 30, 2024 | [Certificate](#)

Posters Presentations

- Spin Estimation of M33 X-8 Black Hole by Spectrum Analysis, PROBE 2025 : **Excellent Grade** for Presentation
- Simulation of the Double-Slit Experiment using Crank–Nicholson Method, PROBE 2024 : **First Prize** for Poster Presentation
- Demonstration of Linear Accelerators using Simple Magnets, JIGYASA 2023
- Construction of Flair Drums, JIGYASA 2022 : **First Prize** for Poster Presentation

Relavant CourseWork

Stellar Physics – *Stellar structure, evolution, pulsations, supernovae* ; **Relativistic Transient Astrophysics** – *Compact objects, binary evolution, accretion, radiative processes* ; **Astrophysics of Galaxies** – *Galactic structure, dynamics, AGN, galaxy evolution* ; **Interstellar & Intergalactic Medium** – *Star formation, chemical evolution, spectroscopy* ; **General Relativity & Cosmology** – *Black holes, Friedman equations, early universe* ; **Observational Astronomy** – *Telescopes, detectors, coordinate systems, multi-wavelength techniques*

Classical & Quantum Mechanics – *Lagrangian/Hamiltonian formalisms, perturbation theory* ; **Mathematical & Statistical Physics** – *PDEs, tensors, special functions, statistical ensembles* ; **Electrodynamics** – *EM wave propagation, radiation, special relativity* ; **Atomic & Molecular Physics** – *Atomic spectra, multi-electron systems, astrophysical applications*

Work Experience

B.K. Birla College (Empowered Autonomous), Kalyan | Assistant Professor (Ad-Hoc)

[Aug 2025 - Present]

- *Taught undergraduate courses in Electrodynamics, Quantum Mechanics, Python Programming & Numerical Methods, and Astronomy & Astrophysics, delivering concept-focused lectures.*
- *Mentored students on multiple short-projects, guiding methodology and analysis.*
- *Support departmental teaching, assessment design, and student academic mentoring.*